Nonnegative and positive factorizations

Connectedness of minimal positive semidefinite factorization orbits

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because topology of optimal PSD factorizations must be controlled beyond size two; community importance concerns nonuniqueness and separated solution families in constrained factorization algorithms.

Area: geometry of constrained matrix factorizations

Resolution — 2026-09-11

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A strictly positive integer 6×6 matrix has ordinary rank six and real positive semidefinite rank three, while its minimal-factor congruence quotient is disconnected. A continuous congruence-invariant orientation takes opposite signs on two explicit factorizations, proving actual disconnectedness in the required quotient topology. Further constructions cover every factor size $k \geq 3$, including strictly positive rational examples by a nonquantitative perturbation argument.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below.

Primary reference: complete authored PDF, standalone TeX, **Theorem 1; Theorem 4 extends the counterexamples to every factor size**. Independent proof review · Authorship and submission record. Verification is independent agent review, not external human peer review or formal certification.

Context and notation

Write $\mathbb{S}_+^k$ for the cone of real symmetric positive semidefinite $k \times k$ matrices. For an entrywise nonnegative matrix $M \in \mathbb{R}_+^{p\times q}$, its **real positive semidefinite rank** is

$$\operatorname{rank}_{\mathrm{psd}}(M) = \min\{k \geq 1 : \exists A_1, \dots, A_p, B_1, \dots, B_q \in \mathbb{S}_+^k, \quad M_{ij} = \operatorname{tr}(A_i B_j) \text{ for all } i, j\}.$$

This definition concerns a family of matrix factors indexed by the rows and columns of M.

Problem statement

Let $k \geq 3$ and $p, q \geq 1$ be integers. Suppose $M \in \mathbb{R}_+^{p\times q}$ satisfies

$$\operatorname{rank}(M) = \frac{k(k+1)}{2}, \qquad \operatorname{rank}_{\mathrm{psd}}(M) = k.$$

Let $\mathcal{F}_k(M)$ be the set of all tuples $(A_1, \dots, A_p, B_1, \dots, B_q)$ in $(\mathbb{S}_+^k)^{p+q}$ satisfying $\operatorname{tr}(A_i B_j) = M_{ij}$ for every i, j. Give it the Euclidean subspace topology. Identify tuples under the changes of basis

$$A_i \longmapsto S^{\mathsf{T}} A_i S, \qquad B_j \longmapsto S^{-1} B_j S^{-\mathsf{T}}, \qquad S \in GL(k, \mathbb{R}).$$

Give the resulting orbit space the quotient topology.

Question

Is $\mathcal{F}_k(M)/GL(k, \mathbb{R})$ connected for every M satisfying these hypotheses?

The ordinary-rank hypothesis is part of the problem. The case $k = 2$ is known to be connected. The question asks whether optimal factors can belong to separated families after changes of basis are identified.

References

Fawzi, Gouveia, Parrilo, Robinson, and Thomas, *Positive semidefinite rank*, §9.2, Problem 9.4. Richard Z. Robinson, *The Positive Semidefinite Rank of Matrices and Polytopes*, University of Washington dissertation (2015), Chapter 7, especially Proposition 7.0.8.

Status check — 2026-09-10

Rechecked Fawzi et al., §9.2, Problem 9.4, including the ordinary-rank hypothesis, and searched for connectedness of PSD factorization orbits. No resolution for k at least three was located. The proved k=2 case is excluded here; disconnected nonnegative factorizations do not automatically remain disconnected in the PSD orbit space. No recent primary reaffirmation of the exact higher-size question was found.